

ESP32C6 Smarthome Final

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Problem Domain:

The current system for managing google home smarthome systems is a bit limited, and is dependent on google servers. For example, if the app were to go down, I would lose a bunch of functions. Also, the app is filled with a huge list of devices, when only a couple are relevant to me.

Solution Domain:

An ESP32C6 sending input to a Raspberry Pi with Homeassistant and the ESPHome addon, which sends API calls to my personal google home system. This means I have instant fine control over the smart devices in my room without having to pick them out. Hosting this on my own Pi also means that this solution is more secure and reliable than using the regular app, since it would continue working if the app were to go down.

Technologies and Tools:

- ESP32C6 with physical components
- Raspberry Pi 3
- Home Assistant (Open source platform for smarthome devices)
- ESPHome (Addon library for ESP devices)
- Wiz Tunable Lighting

Learning With AI: Physical components

In this project, there were several physical components I needed to learn about, including the Arduino and Raspberry Pi. AI helped me learn how to use certain physical components by giving me a quick setup guide so I could try out the component and see how it reacted. It also helped me to decide on and set up the Raspberry Pi with ESPHome, which resulted in a surprise benefit to security and reliability over my original plan.

Learning With AI: Coding

In this project, I had to deal with several new languages including arduino code and YAML script. Learning the documentation for both of these as well as how to combine them was a challenge, but having numerous examples to learn from made it much easier to keep track of. The google home script editor has its own AI helper which can create example automations based on prompts, which was instrumental in getting me started with YAML script.

Video Demo

Youtube Link:

https://youtube.com/shorts/UPkAOc3_4eQ?feature=share

Google Drive Link:

[https://drive.google.com/file/d/14hx9jYhWfONpsX8ZDEQ-GbE8UPL85l4h/view?
usp=drive_link](https://drive.google.com/file/d/14hx9jYhWfONpsX8ZDEQ-GbE8UPL85l4h/view?usp=drive_link)